

MUSKEGON COUNTY AP, MI, USAWMO: **726360**Lat: **43.171N**Lon: **86.237W**Elev: **625**StdP: **14.37**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **14840****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.7	9.0	-2.5	4.9	6.7	1.2	6.0	10.8	27.7	29.9	25.4	30.7	9.4	130	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	86.4	72.1	84.0	70.9	81.8	69.7	75.3	82.7	73.7	80.5	72.3	78.8	11.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	124.8	79.6	71.3	118.4	78.2	69.8	112.3	76.7	39.1	82.9	37.6	80.7	36.3	78.7	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	22.0	19.6	-2.2	90.7	6.3	2.8	-6.8	92.8	-10.5	94.4	-14.0	96.0	-18.6	98.1		
(5)			-2.8	77.7	6.2	2.3	-7.3	79.3	-10.9	80.7	-14.3	82.0	-18.8	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	25.5	26.9	34.8	46.1	57.1	66.7	71.0	69.9	63.0	51.4	40.3	31.2
	DBStd	18.08	9.52	9.51	10.31	8.67	8.32	6.57	5.62	5.15	7.25	8.03	8.11	8.40
	HDD50	3068	758	648	488	174	26	0	0	0	2	79	307	586
	HDD65	6495	1223	1068	938	568	273	56	13	16	122	428	741	1049
	CDD50	2623	0	1	16	58	245	501	651	617	393	123	17	1
	CDD65	574	0	0	1	2	27	107	200	168	63	6	0	0
	CDH74	4197	0	0	6	14	216	827	1564	1124	417	29	0	0
	CDH80	895	0	0	0	0	46	194	379	197	78	1	0	0
Wind	WSAvg	9.4	10.9	10.6	10.1	10.5	9.2	8.1	7.7	7.3	7.9	9.5	10.6	10.6
Precipitation	PrecAvg	33.3	2.4	1.8	2.4	3.1	2.9	2.6	2.4	3.3	3.5	3.2	3.1	2.6
	PrecMax	43.8	5.5	4.6	6.6	7.4	9.6	7.0	6.0	9.9	13.5	9.3	6.8	5.4
	PrecMin	23.1	0.4	0.2	0.4	0.4	0.3	0.5	0.9	0.1	0.2	0.4	0.5	0.8
	PrecStd	5.1	1.1	1.0	1.3	1.3	1.7	1.4	1.1	2.0	2.3	1.8	1.6	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.8	56.3	71.3	76.1	85.2	88.5	90.4	87.6	86.4	77.9	65.1	57.5
		MCWB	50.0	48.9	59.8	60.3	68.3	73.0	75.1	73.5	70.0	67.1	56.5	54.5
	2%	DB	45.4	49.4	61.8	70.4	79.6	85.0	86.7	84.5	82.1	73.0	60.6	50.2
		MCWB	42.4	44.7	53.9	56.6	65.8	70.4	73.0	72.1	69.3	63.8	54.3	47.5
	5%	DB	41.4	43.0	55.5	65.6	75.5	81.9	84.1	82.1	78.8	69.1	56.8	46.3
		MCWB	38.5	38.8	48.4	53.4	63.1	68.5	71.5	70.7	67.8	61.6	51.7	42.9
	10%	DB	38.2	39.1	49.7	61.0	71.7	78.9	81.7	79.9	75.6	64.9	53.1	42.6
		MCWB	35.9	36.1	43.8	50.6	60.4	67.1	70.2	69.5	66.3	58.7	48.3	39.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	51.9	61.2	64.3	71.2	75.4	77.9	77.0	74.1	69.2	59.0	54.5
		MCDB	52.5	53.1	70.0	72.7	81.8	85.6	86.6	84.0	81.5	75.3	62.6	56.8
	2%	WB	43.3	44.5	55.3	59.2	68.1	72.8	75.8	74.9	71.8	66.2	55.3	48.2
		MCDB	45.4	49.4	60.7	66.2	76.3	82.0	83.6	80.8	77.9	70.9	59.3	50.2
	5%	WB	38.8	39.2	49.4	55.2	65.2	70.8	74.1	73.3	69.8	62.5	51.9	43.5
		MCDB	40.8	42.4	54.4	63.3	72.9	78.8	81.2	79.0	75.9	67.9	55.8	46.1
	10%	WB	36.1	36.4	44.1	51.5	62.4	69.0	72.4	71.8	68.0	59.1	48.9	39.4
		MCDB	38.2	39.1	49.5	59.1	69.8	76.2	79.0	77.5	74.2	64.2	53.1	42.6
Mean Daily Temperature Range	5% DB	MDBR	10.6	12.3	15.2	17.9	18.8	18.2	17.5	16.8	18.2	15.7	12.3	9.7
		MCDBR	14.1	16.5	21.5	24.7	22.3	21.4	18.7	18.2	20.4	20.7	16.9	14.3
		MCWBR	12.7	13.0	15.0	14.6	12.4	10.5	9.1	9.0	10.3	13.2	12.9	12.9
	5% WB	MCDBR	13.4	15.5	19.8	21.4	19.6	17.8	15.9	14.3	16.1	18.0	15.1	13.7
		MCWBR	12.5	12.9	15.3	14.9	12.7	10.0	8.7	8.2	9.8	13.3	13.0	13.2
Clear-Sky Solar Irradiance	taub	0.299	0.328	0.342	0.372	0.403	0.418	0.418	0.414	0.390	0.353	0.319	0.298	
	taud	2.407	2.308	2.368	2.314	2.294	2.304	2.306	2.321	2.356	2.445	2.482	2.453	
	Ebn at Noon	265	273	285	283	276	272	270	267	265	262	256	255	
	Edh at Noon	24	32	34	39	41	41	40	38	34	28	23	22	
All-Sky Solar Radiation	RadAvg	380	643	1130	1506	1770	1981	2015	1715	1364	840	486	308	
	RadStd	43	81	110	139	161	127	95	103	116	102	72	45	